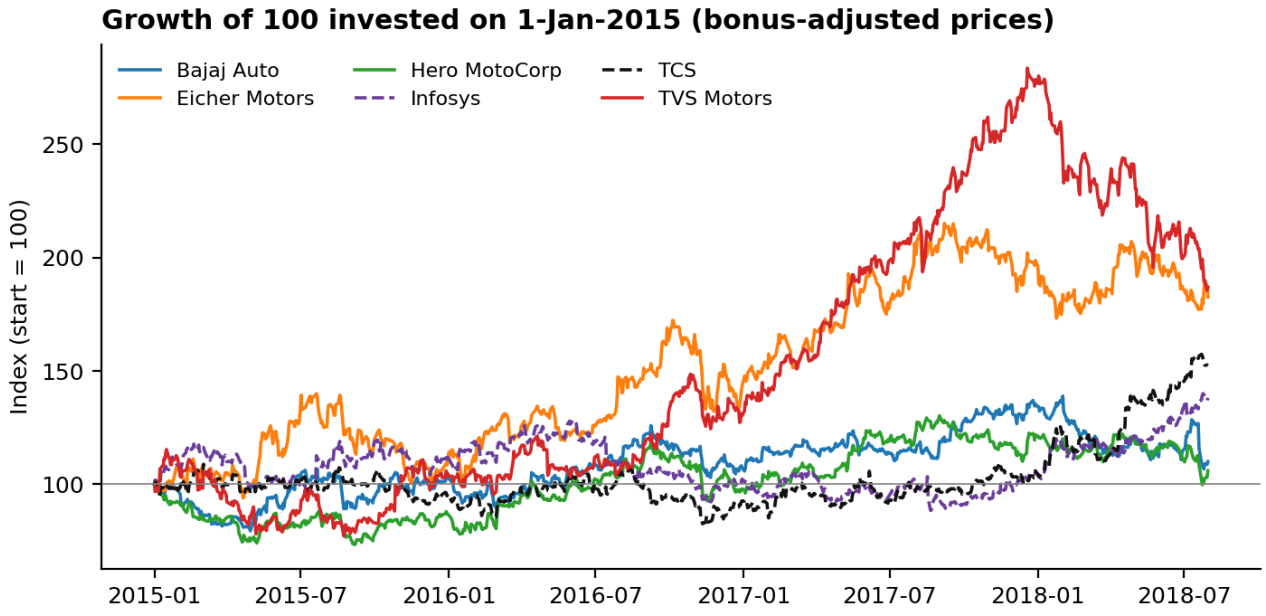
Auto vs IT Stocks: What 3.5 Years of Data Say

SQL-based analysis of Bajaj Auto, Eicher Motors, Hero MotoCorp, TVS Motors, Infosys and TCS | 1 Jan 2015 - 31 Jul 2018 | 889 trading days each

Executive summary

- **Winners:** TVS Motors (+86.9%) and Eicher Motors (+82.6%) led, followed by TCS (+52.4%) and Infosys (+38.2%). Bajaj Auto (+10.0%) and Hero MotoCorp (+6.0%) returned only 2.7% and 1.6% a year (CAGR).
- **Higher return came with higher risk:** TVS and Eicher were the most volatile (30.8% and 29.1% annualised) and TVS had the deepest fall (-34.6%). TCS was the calmest (21.6%) and delivered nearly the best return per unit of risk (only Eicher and TVS scored slightly higher).
- **2018 was a sector rotation:** in Jan-Jul 2018 TCS (+43.8%) and Infosys (+31.3%) surged while TVS (-32.9%), Bajaj (-18.7%), Hero (-13.0%) and Eicher (-8.2%) fell. The Auto basket peaked at +83.7% by end-2017 and then gave back most of its lead.
- **Diversification works:** Auto and IT stocks are only weakly correlated (average 0.14), versus 0.35 inside Auto and 0.41 between Infosys and TCS.
- **Volume signals conviction:** on days when volume was more than twice its 20-day average, prices moved about 1.4x to 2x more than on normal days for every stock.
- **Data catch:** the raw files are *not* adjusted for the 1:1 bonus issues of Infosys (15 Jun 2015) and TCS (31 May 2018). Unadjusted, both would show a fake ~50% crash. All numbers in this report are adjusted.



1. Data, cleaning and method

Data. Six daily files from the stock exchange with open, high, low, close, WAP, shares traded, number of trades, turnover, deliverable quantity and delivery %. Each file has 889 rows from 1 Jan 2015 to 31 Jul 2018. Four stocks are Auto (Bajaj, Eicher, Hero, TVS) and two are IT (Infosys, TCS).

Method. The CSVs were loaded into MySQL, merged into one table with a stock and sector column, checked for quality, adjusted for corporate actions, and analysed using window functions (LAG, rolling AVG, RANK, FIRST_VALUE). The full code is in `stock_analysis.sql`.

Data-quality checks and fixes

Check	Result	Action
Duplicate dates	0 in all six files	None needed
Invalid prices (close <= 0, high < low)	0 rows	None needed
Missing values	1 row per stock, in Deliverable Qty and Delivery % only	Kept as NULL; ignored in averages (not set to 0)
Moves above 25% in one day	Infosys 15-Jun-2015 (-49.9%), TCS 31-May-2018 (-50.4%)	Identified as ex-bonus/split dates
After adjustment	No move above 25% remains	Verified with the same query; prices before the ex-date divided by 2, share volume multi

Why the adjustment matters

Without it, TCS would show a total return of -23.8% instead of +52.4%, and Infosys -30.9% instead of +38.2% (sign flips!). Volatility and maximum drawdown would also be inflated by a "crash" that never happened to shareholders.

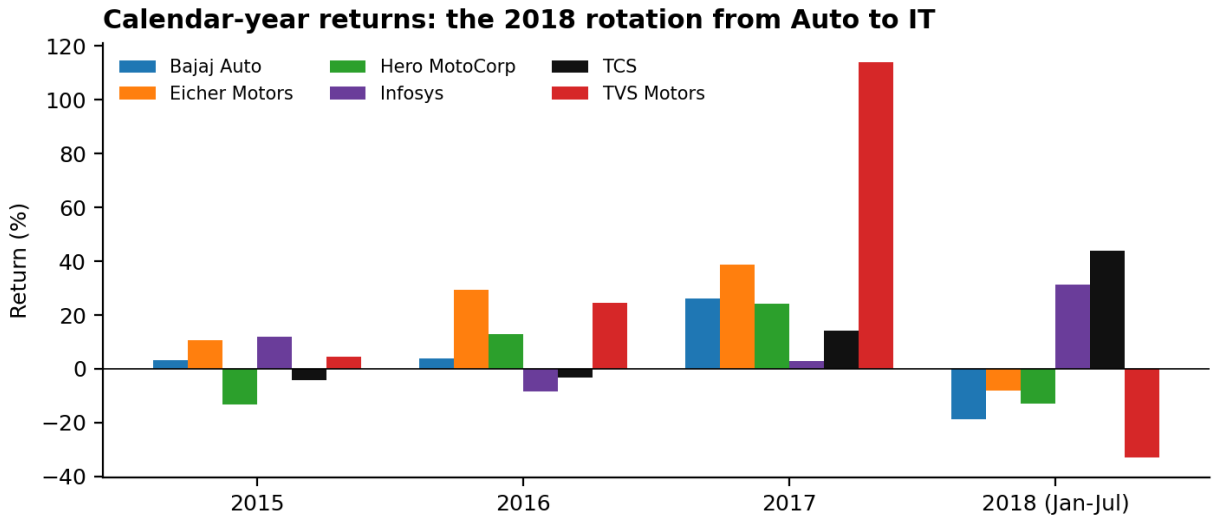
Definitions

- **Daily return** = today close / yesterday close - 1. **Volatility** = standard deviation of daily returns, annualised by multiplying by the square root of 252.
- **Max drawdown** = biggest fall from a running peak to a later low. **Return-to-risk** = total return divided by annualised volatility.
- **Delivery %** = share of traded volume actually taken for delivery; a high value suggests investors, a low value suggests short-term traders.

2. Returns: who made money?

Stock	Sector	Start close	End close	Total return	CAGR	Rank
TVS Motors	Auto	276.85	517.45	+86.9%	19.1%	1
Eicher Motors	Auto	15,239.15	27,820.95	+82.6%	18.3%	2
TCS	IT	1,274.10*	1,941.25	+52.4%	12.5%	3
Infosys	IT	987.90*	1,365.00	+38.2%	9.5%	4
Bajaj Auto	Auto	2,454.10	2,700.70	+10.0%	2.7%	5
Hero MotoCorp	Auto	3,107.30	3,293.80	+6.0%	1.6%	6

* adjusted for the 1:1 bonus issue. Rs per share.



Calendar-year returns

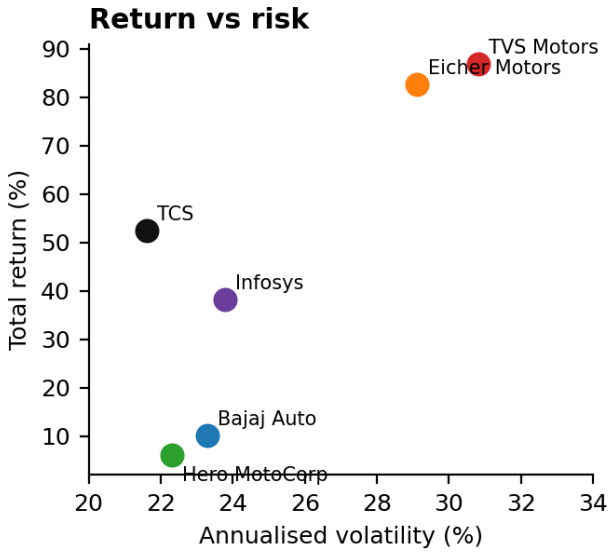
Stock	2015	2016	2017	2018 (Jan-Jul)
Bajaj Auto	3.2%	4.0%	26.2%	-18.7%
Eicher Motors	10.6%	29.4%	38.9%	-8.2%
Hero MotoCorp	-13.3%	13.0%	24.4%	-13.0%
TVS Motors	4.6%	24.5%	113.9%	-32.9%
Infosys	11.8%	-8.5%	2.8%	31.3%
TCS	-4.3%	-3.2%	14.3%	43.8%

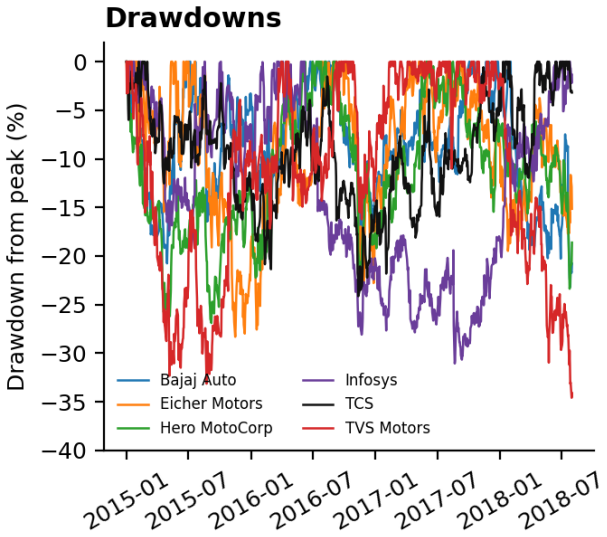
- **TVS had one extraordinary year:** +113.9% in 2017 more than doubled the stock, then it lost a third in 2018. Most of its 3.5-year gain came from that single year.
- **Eicher was the steadier compounder:** positive in 2015, 2016 and 2017, and only -8.2% in the 2018 sell-off.
- **IT lagged for two years and then caught up:** an equal-weight IT basket was at 105.6 at end-2017 (vs 183.7 for Auto) but 145.3 by July 2018, almost level with Auto at 146.4.

3. Risk: what did it cost to earn those returns?

Stock	Annual volatility	Worst day	Best day	Max drawdown	Trough date	Return / risk
TCS	21.6%	-5.2%	+6.8%	-24.1%	11 Nov 2016	2.42
Hero MotoCorp	22.3%	-6.2%	+6.7%	-26.9%	7 Sep 2015	0.27
Bajaj Auto	23.3%	-9.1%	+7.4%	-23.2%	26 Jul 2018	0.43
Infosys	23.8%	-9.6%	+11.1%	-31.1%	21 Aug 2017	1.60
Eicher Motors	29.1%	-9.3%	+7.0%	-28.3%	16 Nov 2015	2.84
TVS Motors	30.8%	-9.7%	+13.2%	-34.6%	30 Jul 2018	2.82

Sorted from calmest to most volatile. Return / risk = total return / annualised volatility.

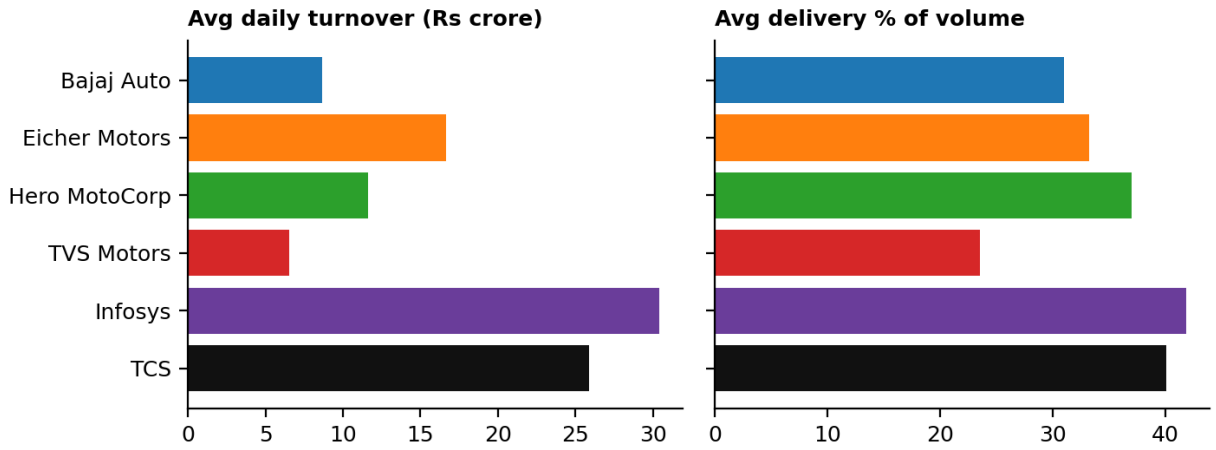




Key points

- **TVS Motors is the high-risk, high-reward stock:** highest volatility, biggest drawdown (-34.6%, still going on at the end of the data) and the best single day (+13.2%).
- **Hero and Bajaj were the worst bargain:** they were about as risky as TCS (22-23% volatility) but returned a fraction of it. Their return-to-risk scores (0.27 and 0.43) are the lowest.
- **Eicher gave the best payoff for the risk taken** (2.84), narrowly ahead of TVS (2.82), though both with big swings. TCS (2.42) delivered nearly as much with the lowest volatility.
- **Event risk shows up in IT too:** Infosys fell 9.6% in one day on 18 Aug 2017 (the day its CEO resigned) and was 31% below its peak by 21 Aug 2017; it later recovered fully.
- **Trend status at 31 Jul 2018:** the 20-day average was above the 50-day (bullish) for TCS, Infosys and Bajaj, and below (bearish) for TVS, Eicher and Hero.

4. How the stocks trade: volume, delivery and daily patterns

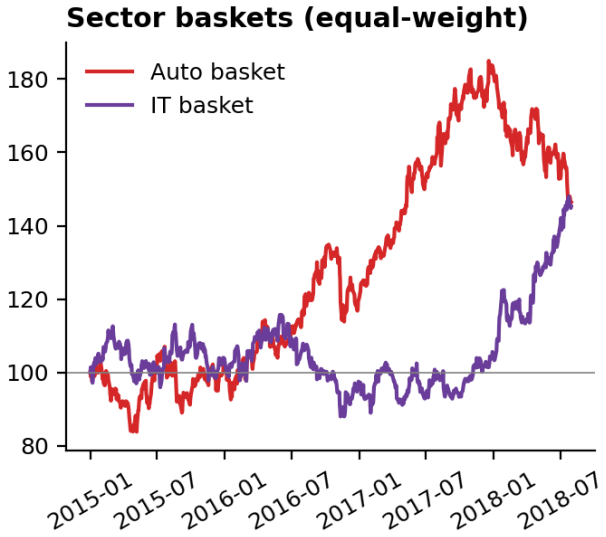


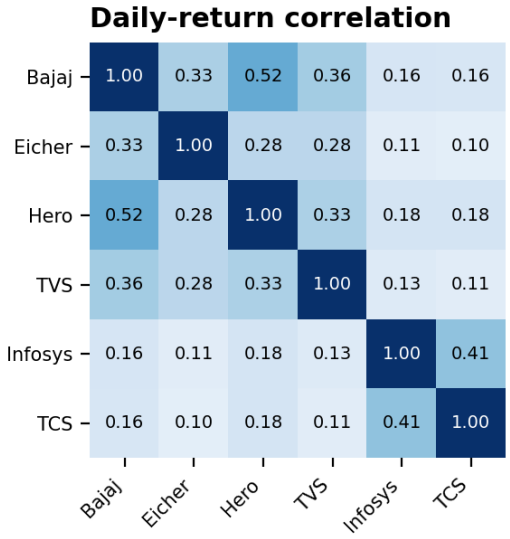
Stock	Avg trades / day	Avg trade size (Rs 000)	Avg intraday range	Avg volume-spike move	Normal-day move
Infosys	7,742	42	2.11%	1.99%	0.98%
TCS	5,024	55	1.93%	1.75%	0.89%
TVS Motors	3,447	25	3.00%	2.60%	1.27%
Hero MotoCorp	2,329	52	2.17%	1.41%	1.00%
Bajaj Auto	2,010	49	2.20%	1.89%	0.96%
Eicher Motors	1,669	91	2.59%	2.14%	1.27%

Volume spike = shares traded above 2x the previous 20-day average. Moves shown are the average absolute daily % change.

- **IT stocks are the most liquid** (Rs 26-30 crore a day) and have the highest delivery share (40-42%): more investor-driven. **TVS has the lowest delivery (23.6%)** and the highest intraday range (3.0%), pointing to heavier short-term trading, consistent with its high volatility.
- **Volume confirms price moves:** on spike days TVS moved 2.6% vs 1.3% normally; Infosys 2.0% vs 1.0%; TCS 1.8% vs 0.9%. A sudden volume jump is a useful warning that a larger move is under way.
- **Delivery % does not separate up days from down days:** differences are 0.4 to 2 points, too small to act on.
- **Intraday pattern:** every stock gapped up on average at the open (+0.08% to +0.28% vs previous close) but closed above the open on only 43-47% of days (average open-to-close -0.02% to -0.20%). Early-morning optimism tended to fade by the close.
- **Calendar effects are weak.** May and October were the best months, November the worst (average daily return negative for both sectors), and individual weekday effects (e.g. Friday for TVS) are small, but with only 3-4 years of data these should be treated as curiosities, not trading rules.

5. Sector view and diversification





Measure	Auto (4 stocks)	IT (2 stocks)
Average daily return	0.053%	0.052%
Daily volatility	1.68%	1.43%
Avg daily turnover per stock	Rs 10.9 crore	Rs 28.2 crore
Avg correlation within sector	0.35	0.41 (Infosys-TCS)
Basket value 31 Dec 2017 (start = 100)	183.7	105.6
Basket value 31 Jul 2018	146.4	145.3

- **Same average daily return, different path:** Auto earned its gains early and with higher volatility; IT earned them late and more smoothly.
- **The most similar pair is Bajaj-Hero (0.52)**, both motorcycle makers. Eicher is the most independent auto stock. Auto-IT pairs sit at 0.10-0.18, so holding both sectors would have smoothed the ride (the 2018 rotation is the example).

Conclusions and recommendations

- **Best all-round (return and stability):** TCS, then Eicher Motors. **Highest-risk, high-reward:** TVS Motors. **Weakest risk-reward in the period:** Hero MotoCorp and Bajaj Auto.
- **Blend sectors:** the low Auto-IT correlation is the strongest, most reliable finding in this data set.
- **Use volume as a risk flag:** watch for volume above 2x average, which has coincided with moves of roughly double the normal size.
- **Always adjust for bonus and split events** before comparing returns; the unadjusted data would have reversed the conclusion for TCS and Infosys.

Limitations

Only 3.6 years and six stocks, ending July 2018; no dividends are included (price return only); the TCS and Infosys bonus adjustments were identified from the price data and the 1:1 ratio was inferred from the near-50% drop (it matches the public 1:1 bonus announcements, but should be verified against exchange records). Results describe the past and are not investment advice.